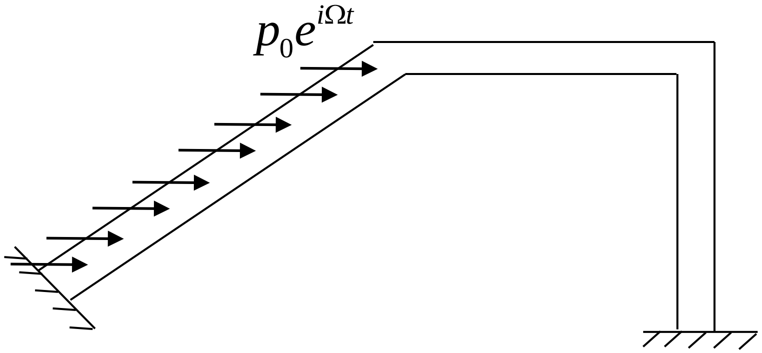
The Finite Element Method (FEM) in the study of continuum dynamics

Aim and justification of the approach

Analytical (“exact”) solutions for continuum mechanics are:

- limited to simple cases (e.g. axial, torsional, bending wibrations of simple beam systems)

- even in these cases, the use of analytical solutions is oten impractical and lacks generality



The Finite Element Method allows for:

- An approximate but very general formulation

- Easy implementation in a computer code.

How does the FEM work ?

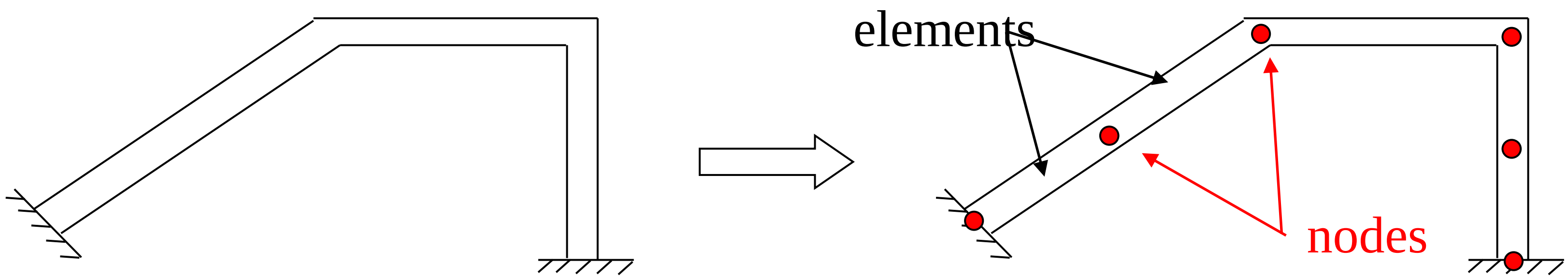
The Finite Element method is a **discretisation** technique, that allows to describe the motion of a distributed parameter system in terms of a discrete number *n* of degrees of freedom. This obviously entails some approximation, that may however be kept within acceptable limits

The discretisation is based on the following process:

1)The structure is divided into a finite number of parts, called “finite elements”

2) in each finite element a finite number of points / sections are identified and called “nodal sections” or “nodes”

3)The motion of all other sections in the finite element is then expressed as function of the motion of the nodal sections, so that the unknowns of the problem become the displacements of the nodes



Advantages of the Finite Element method

The main advantages of the Finite Element methods may be summarised as follows:

1) The method is general and widely applicable
2) It may be coded into an automated computational procedure
3) Mathematical libraries are available for the efficient numerical solution of continuum dynamics problems by the Finite Element method.
4) The FEM allows for a full and easy integration with CAD software for both establishing the model and for displaying the results of analyses.

Contents of the lesson

The lesson will be confined to the study of plane beam systems undergoing axial and bending vibrations

The following study will be performed:

–Free and forced motion

- Forcing produced by concentrated and/or distributed loads
- Forcing produced by displacements in the constraints

The following topics will otherwise not be trated:

- spatial beam systems
- study of continuum systems different from beams, which means 2D and 3D continua

Outline of the FEM application

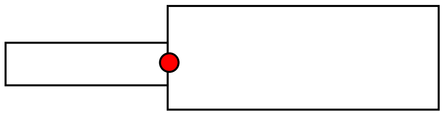
1) The continuum system is divided into finite portions (meshing)
2) Choice of “global” and “local” references
3) Removal of constraints and introduction of the constraint forces
4) Shape functions
5) Expression of energetic functions for the element in its own local reference
6) Coordinate transformation from local to global reference
7) Numbering of the nodal coordinates and structural matrices assembling
8) Model of structural damping
9) Re-introduction of constraints, partition of matrices and final form of the system’s equations of motion
10) Solution of the equations of motion

Step 1: subdivision into finite elements

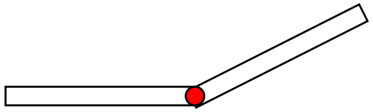
The division of the system into finite elements (meshing) is based (for beams systems) on the following two criteria:

1) Introduce a nodal section in presence of any of the following "discontinuities"

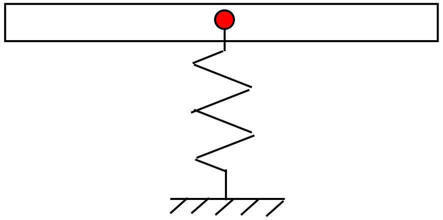
- Variation of beam section properties



- Intersection of 2 (or more) beams with different axis direction



- presence of a lumped element (spring / mass / damper)



Step 1: subdivision into finite elements

2) Furthermore, the length of the finite elements should not exceed a limit value. In fact:

- It will be shown that the approximation of the system motion is the EXACT solution as far as STATIC loads applied on the nodes

-In the case of dynamic loads, in order to get a good approximation of the actual displacements of the system, each finite element has to work in the QUASI-STATIC field, which means at freqeuncies well below its first resonance:

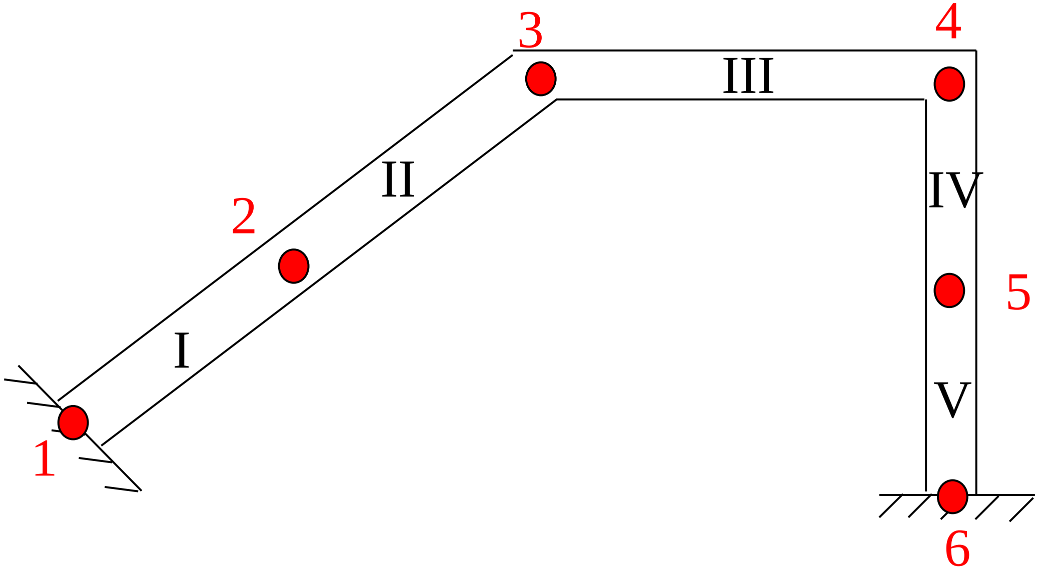
$$\Omega_{\max} << \omega_k = \left(\frac{\pi}{L_k}\right)^2 \sqrt{\frac{EJ_k}{m_k}} \Rightarrow L_k < L_{\max}$$

Ω_{max} = Maximum circular frequency of time dependent forces (frequency range of validity of the model)

ω_k = lowest natural frequency of the k-th finite element

Step 1: subdivision into finite elements

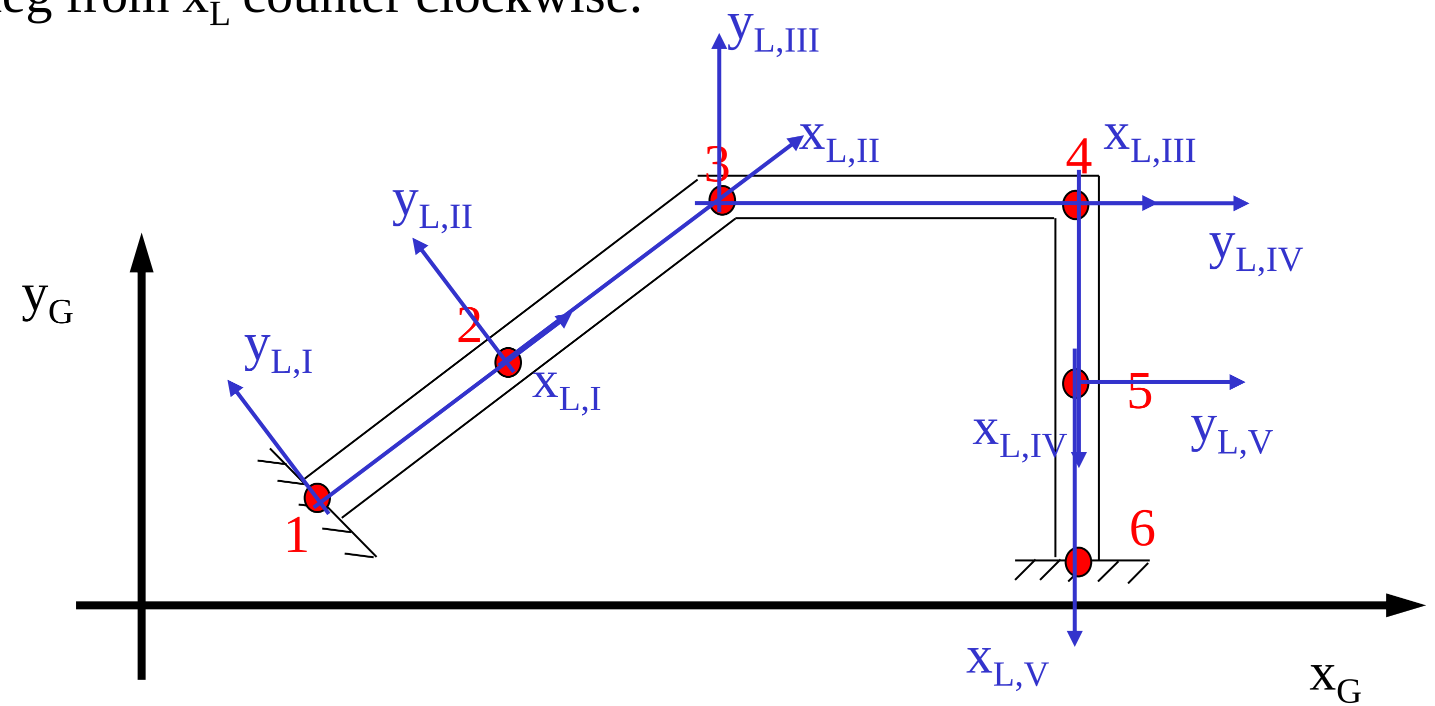
As an example of the procedure, a possible mesh for a simple beam system is shown below, resulting in 6 nodes and 5 finite elements



In the following, nodes are conventionally numbered by arabic numbers (from 1 to 6), while finite elements are numbered by roman numbers (from I to V). The numbering of nodes and elements is purely conventional and does not bear any implication on the results of the calculation.

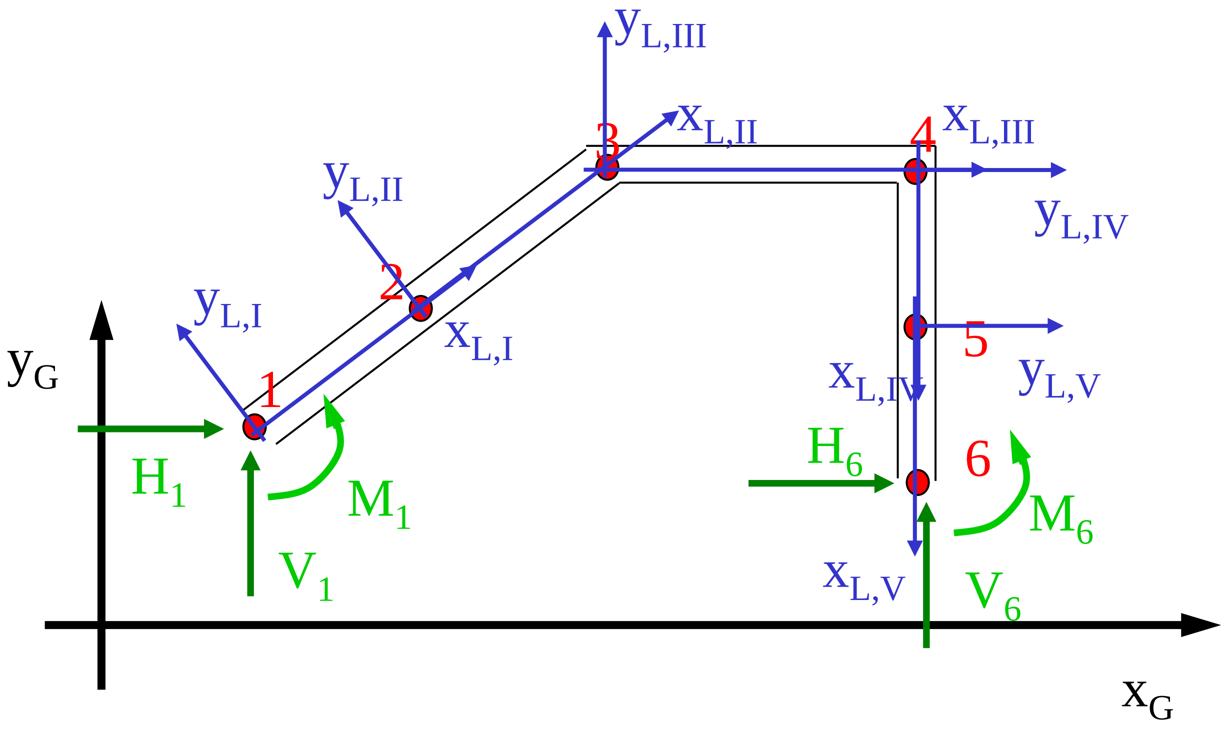
Step 2: Global and local references

After the system discretisation, a global reference (axes x_G,y_G) is defined for the whole system, while for each beam element a local reference (axes x_L,y_L) is taken, where the x_L axis is parallel to the beam element axis and y_L is rotated 90 deg from x_L counter clockwise.



Step 3: removal of constraints and introduction of constraint forces

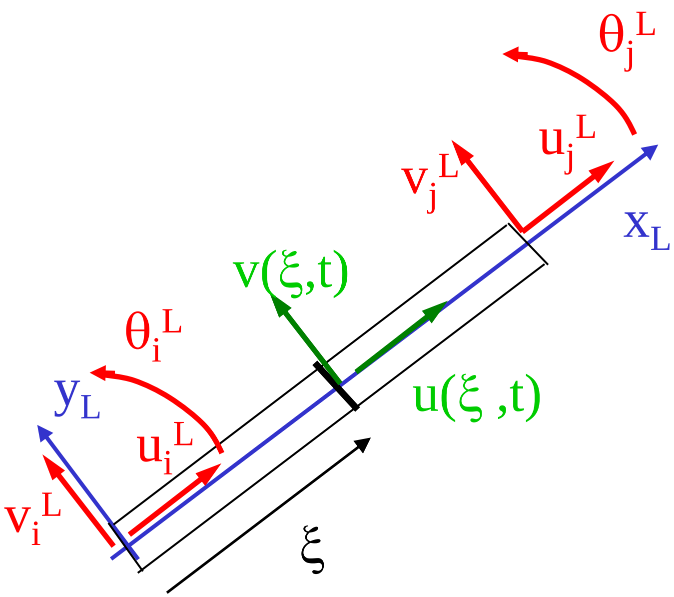
To treat generally the presence of any combination of constraints on the system, constraints are initially "removed" from the system, by introducing in their place the (unknown) constraint forces. The presence of the constraints will be recovered at the end of the solution procedure (step 9)



Step 4: Shape functions – axial motion

-Shape functons are used to express the motion of all sections in the beam element as function of the nodal coordinates

-Shape functions are defined in the *local* reference



Axial motion:

$$u(\xi,t) = a + b\xi$$

By equating the above expression to the nodal displacements in the nodes:

$$u(0,t) = u_i^L \quad ; \quad u(L_k,t) = u_j^L$$

$$u(\xi,t) = \left(1 - \frac{\xi}{L_k}\right) u_i^L(t) + \left(\frac{\xi}{L_k}\right) u_j^L(t)$$

Step 4: Shape functions – axial motion

The expression of axial displacements in the beam element may be re-formulated in a matrix form through the following definition:

$$\underline{x}_k^L = \begin{Bmatrix} u_i^L \\ v_i^L \\ \vartheta_i^L \\ u_j^L \\ v_j^L \\ \vartheta_j^L \end{Bmatrix}$$

Is the 6-element vector collecting the element nodal coordinates expressed along the local reference (local coordinates):

$$\underline{f}_u(\xi) = \begin{Bmatrix} (1-\frac{\xi}{L_k}) \\ 0 \\ 0 \\ \frac{\xi}{L_k} \\ 0 \\ 0 \end{Bmatrix}$$

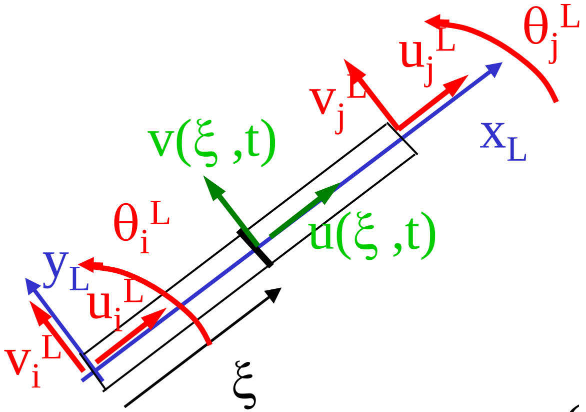
Is the axial shape functions vector, allowing to express the axial movements as:

$$u(\xi,t) = \underline{f}_u(\xi)^T \, \underline{x}_k^L(t)$$

Step 4: Shape functions – bending motion

The bending motion of the beam w(ξ,t) is approximated along the beam element by a third order function:

$$v(\xi,t) = a + b\xi + c\xi^2 + d\xi^3$$



By equating the displacements and rotations according to the above expression with the nodal displacements and rotations:

$$v(0,t) = v_i^L \; ; v(L_k,t) = v_j^L \; ; \left.\frac{\partial v}{\partial \xi}\right|_{\xi=0} = \vartheta_i^L \; ; \left.\frac{\partial v}{\partial \xi}\right|_{\xi=L_k} = \vartheta_j^L$$

$$\Downarrow$$

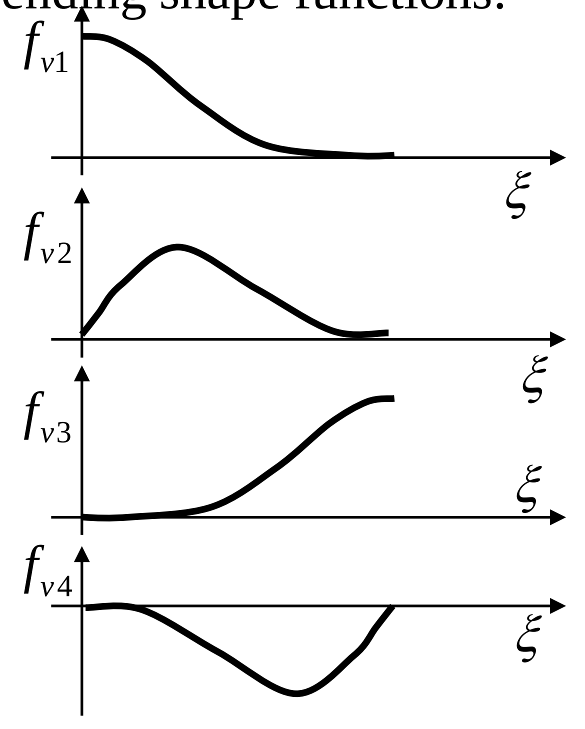
$$v(\xi,t) = f_{V1}(\xi) y_i^L(t) + f_{V2}(\xi)\vartheta_i^L(t) + f_{V3}(\xi) y_j^L(t) + f_{V4}(\xi)\vartheta_j^L(t)$$

The bending motion is expressed as a linear combination of the nodal coordinates in the local reference, f_{v1}, f_{v2}, f_{v3} and f_{v4} being the “shape functions” in this case cubic functions of ξ.

Step 4: Shape functions – bending motion

A matrix expression for the bending displacement of the beam element is obtained by defining the following vector of bending shape functions:

$$\underline{f}_v(\xi) = \begin{Bmatrix} 0 \\ f_{v1}(\xi) \\ f_{v2}(\xi) \\ 0 \\ f_{v3}(\xi) \\ f_{v4}(\xi) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 2\left(\frac{\xi}{L_k}\right)^3 - 3\left(\frac{\xi}{L_k}\right)^2 + 1 \\ L_k\left[\left(\frac{\xi}{L_k}\right)^3 - 2\left(\frac{\xi}{L_k}\right)^2 + \frac{\xi}{L_k}\right] \\ 0 \\ -2\left(\frac{\xi}{L_k}\right)^3 + 3\left(\frac{\xi}{L_k}\right)^2 \\ L_k\left[\left(\frac{\xi}{L_k}\right)^3 - \left(\frac{\xi}{L_k}\right)^2\right] \end{Bmatrix}$$

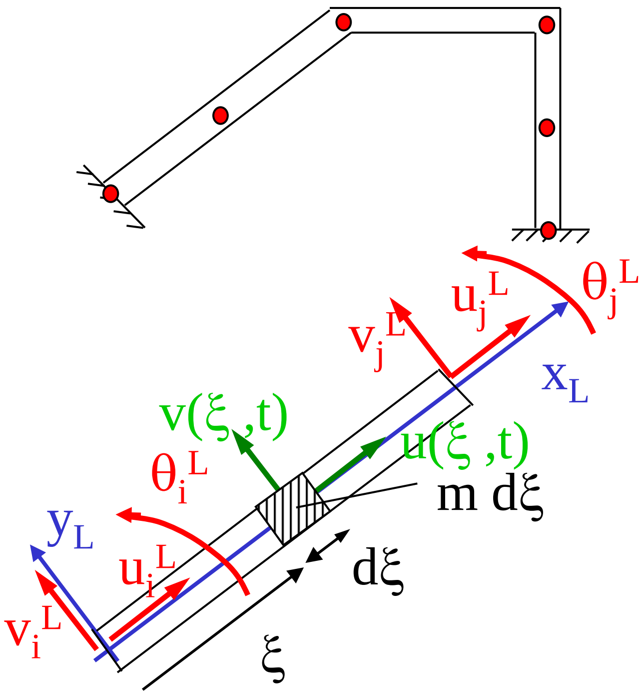


And then:

$$v(\xi,t) = \underline{f}_v(\xi)^T \underline{x}_k^L(t)$$

Step 5: Expression of energetic quantities

Kinetic energy



$$T=\sum_k T_k\ ;\ T_k=\tfrac{1}{2}\int_0^{L_k} m\left[\left(\tfrac{\partial u}{\partial t}\right)^2+\left(\tfrac{\partial v}{\partial t}\right)^2\right]d\xi=$$

$$=\tfrac{1}{2}\int_0^{L_k}\left(\tfrac{\partial u}{\partial t}\right)^T m\left(\tfrac{\partial u}{\partial t}\right)d\xi+\tfrac{1}{2}\int_0^{L_k}\left(\tfrac{\partial v}{\partial t}\right)^T m\left(\tfrac{\partial v}{\partial t}\right)d\xi$$

Considering the expression of axial and bending motion as function of nodal displacements via the shape functions:

$$u(\xi,t)=\underline{f}_u(\xi)^T\underline{x}_k^L(t)\Rightarrow\tfrac{\partial u}{\partial t}=\underline{f}_u(\xi)^T\underline{\dot{x}}_k^L(t)$$

$$v(\xi,t)=\underline{f}_v(\xi)^T\underline{x}_k^L(t)\Rightarrow\tfrac{\partial v}{\partial t}=\underline{f}_v(\xi)^T\underline{\dot{x}}_k^L(t)$$

$$T_k=\tfrac{1}{2}\left(\underline{\dot{x}}_k^L\right)^T\underbrace{\int_0^{L_k}\underline{f}_u(\xi)m\underline{f}_u(\xi)^T d\xi}_{[m_u]}\left(\underline{\dot{x}}_k^L\right)+\tfrac{1}{2}\left(\underline{\dot{x}}_k^L\right)^T\underbrace{\int_0^{L_k}\underline{f}_v(\xi)m\underline{f}_v(\xi)^T d\xi}_{[m_v]}\left(\underline{\dot{x}}_k^L\right)=$$

$$=\tfrac{1}{2}\left(\underline{\dot{x}}_k^L\right)^T\left([m_u]+[m_v]\right)\left(\underline{\dot{x}}_k^L\right)=\tfrac{1}{2}\left(\underline{\dot{x}}_k^L\right)^T\left[m_k^L\right]\left(\underline{\dot{x}}_k^L\right)$$

Step 5: Expression of energetic quantities
Mass matrix of the beam finite element

$$\left[m_k^L\right]=\begin{bmatrix}
\frac{1}{3}mL_k & 0 & 0 & \frac{1}{6}mL_k & 0 & 0\\
0 & \frac{13}{35}mL_k & \frac{-11}{210}mL_k^2 & 0 & \frac{9}{70}mL_k & \frac{13}{420}mL_k^2\\
0 & \frac{-11}{210}mL_k^2 & \frac{1}{105}mL_k^3 & 0 & \frac{-13}{420}mL_k^2 & \frac{-1}{140}mL_k^3\\
\frac{1}{6}mL_k & 0 & 0 & \frac{1}{3}mL_k & 0 & 0\\
0 & \frac{9}{70}mL_k & \frac{-13}{420}mL_k^2 & 0 & \frac{13}{35}mL_k & \frac{11}{210}mL_k^2\\
0 & \frac{13}{420}mL_k^2 & \frac{-1}{140}mL_k^3 & 0 & \frac{11}{210}mL_k^2 & \frac{1}{105}mL_k^3
\end{bmatrix}$$

Step 5: Expression of energetic quantities
Potential energy

$$V=\sum V_k\ ;\ V_k=\underbrace{\tfrac{1}{2}\int_0^{L_k} EA_k\left(\tfrac{\partial u}{\partial \xi}\right)^2 d\xi}_{\text{Contribution of axial def.}}+\underbrace{\tfrac{1}{2}\int_0^{L_k} EJ_k\left(\tfrac{\partial^2 v}{\partial \xi^2}\right)^2 d\xi}_{\text{Contribution of bending def.}}=$$

$$=\tfrac{1}{2}\int_0^{L_k}\left(\tfrac{\partial u}{\partial \xi}\right)^T EA_k\left(\tfrac{\partial u}{\partial \xi}\right)d\xi+\tfrac{1}{2}\int_0^{L_k}\left(\tfrac{\partial^2 v}{\partial \xi^2}\right)^T EJ_k\left(\tfrac{\partial^2 v}{\partial \xi^2}\right)d\xi$$

Substituting the expression of axial and bending motion as function of nodal displacements via the shape functions:

$$u(\xi,t)=\underline{f}_u(\xi)^T\underline{x}_k^L(t)\Rightarrow\tfrac{\partial u}{\partial \xi}=\underline{f}'_u(\xi)^T\underline{x}_k^L(t)$$

$$v(\xi,t)=\underline{f}_v(\xi)^T\underline{x}_k^L(t)\Rightarrow\tfrac{\partial^2 v}{\partial \xi^2}=\underline{f}''_v(\xi)^T\underline{x}_k^L(t)$$

$$V_k=\tfrac{1}{2}\left(\underline{x}_k^L\right)^T\underbrace{\int_0^{L_k}\underline{f}'_u(\xi)EA_k\underline{f}'_u(\xi)^T d\xi}_{[k_u]}\left(\underline{x}_k^L\right)+\tfrac{1}{2}\left(\underline{x}_k^L\right)^T\underbrace{\int_0^{L_k}\underline{f}''_v(\xi)EJ_k\underline{f}''_v(\xi)^T d\xi}_{[k_v]}\left(\underline{x}_k^L\right)=$$

$$=\tfrac{1}{2}\left(\underline{x}_k^L\right)^T\left[k_k^L\right]\left(\underline{x}_k^L\right)$$

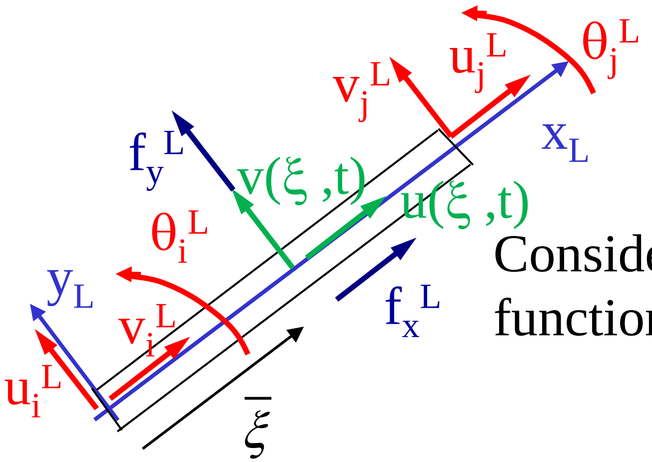
Step 5: Expression of energetic quantities
Stiffness matrix of the beam finite element

$$\left[k_k^L\right]=\begin{bmatrix} \frac{EA}{L_k} & 0 & 0 & -\frac{EA}{L_k} & 0 & 0 \\ 0 & \frac{12EJ_y}{L_k^3} & \frac{-6EJ_y}{L_k^2} & 0 & \frac{-12EJ_y}{L_k^3} & \frac{-6EJ_y}{L_k^2} \\ 0 & \frac{-6EJ_y}{L_k^2} & \frac{4EJ_y}{L_k} & 0 & \frac{6EJ_y}{L_k^2} & \frac{2EJ_y}{L_k} \\ -\frac{EA}{L_k} & 0 & 0 & \frac{EA}{L_k} & 0 & 0 \\ 0 & \frac{-12EJ_y}{L_k^3} & \frac{6EJ_y}{L_k^2} & 0 & \frac{12EJ_y}{L_k^3} & \frac{6EJ_y}{L_k^2} \\ 0 & \frac{-6EJ_y}{L_k^2} & \frac{2EJ_y}{L_k} & 0 & \frac{6EJ_y}{L_k^2} & \frac{4EJ_y}{L_k} \end{bmatrix}$$

Step 5: Expression of energetic quantities Virtual work of time dependent forces

$$\delta L = \delta L_{ext} + \delta L_R$$

Case 1: Concentrated force



$$\delta L_k = \left(\delta u(\bar{\xi})\right)^T f_x^L + \left(\delta v(\bar{\xi})\right)^T f_y^L$$

Considering the expression of axial and bending motion as function of nodal displacements via the shape functions:

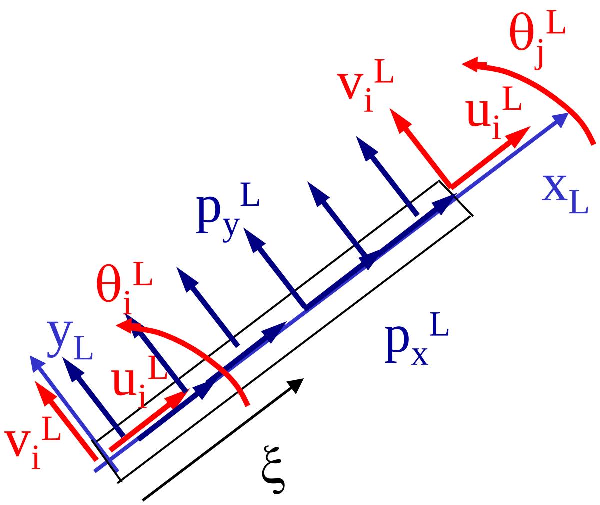
$$u(\xi,t) = \underline{f}_u(\xi)^T \underline{x}_k^L(t) \Rightarrow \delta u = \underline{f}_u(\xi)^T \delta \underline{x}_k^L$$

$$v(\xi,t) = \underline{f}_v(\xi)^T \underline{x}_k^L(t) \Rightarrow \delta v = \underline{f}_v(\xi)^T \delta \underline{x}_k^L$$

$$\delta L_k = \left(\delta \underline{x}_k^L\right)^T \underbrace{\left(\underline{f}_u(\bar{\xi}) f_x^L + \underline{f}_v(\bar{\xi}) f_y^L\right)}_{\underline{F}_k^L} = \left(\delta \underline{x}_k^L\right)^T \underline{f}_k^L$$

Step 5: Expression of energetic quantities
Virtual work of time dependent forces

Case 2: Distributed force



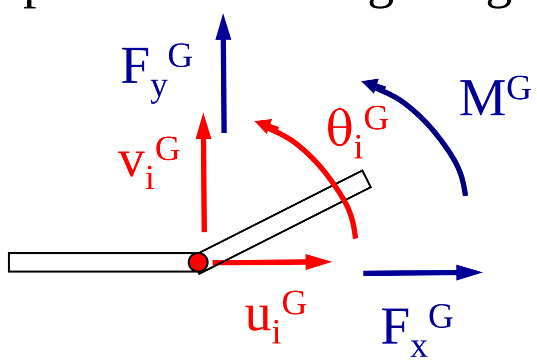
$$\delta L_k = \int_0^{L_k} \left(\delta u(\xi)\right)^T p_x^L(\xi) d\xi + \int_0^{L_k} \left(\delta v(\xi)\right)^T p_y^L(\xi) d\xi =$$

$$= \left(\delta \underline{x}_k^L\right)^T \underbrace{\int_0^{L_k} \underline{f}_u(\xi) p_x^L(\xi) d\xi + \int_0^{L_k} \underline{f}_v(\xi) p_y^L(\xi) d\xi}_{\underline{F}_k^L} =$$

$$= \left(\delta \underline{x}_k^L\right)^T \underline{f}_k^L$$

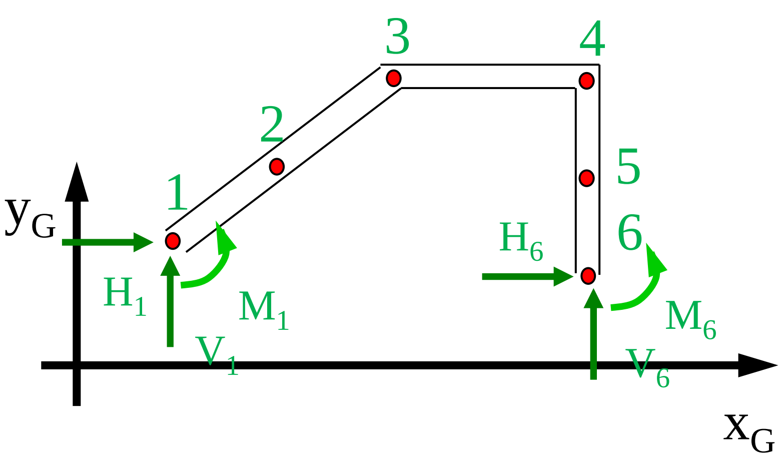
Step 5: Expression of energetic quantities
Virtual work of time dependent forces

Case 1b: concentrated force on a node: since nodal displcaments are the independent coordinates of the problem, it is simpler in this case to use directly the nodal displacements along the global reference



$$\delta L_k = \delta u_i^G F_x^G + \delta v_i^G F_y^G + \delta \vartheta_i^G M^G = \left(\delta \underline{x}_i^G\right)^T \begin{Bmatrix} F_x^G \\ F_y^G \\ M^G \end{Bmatrix}$$

The same happens for the virtual work of the constraint forces:



$$\delta L_R = \delta u_1^G H_1^G + \delta v_1^G V_1^G + \ldots + \delta \vartheta_6^G M_6^G =$$

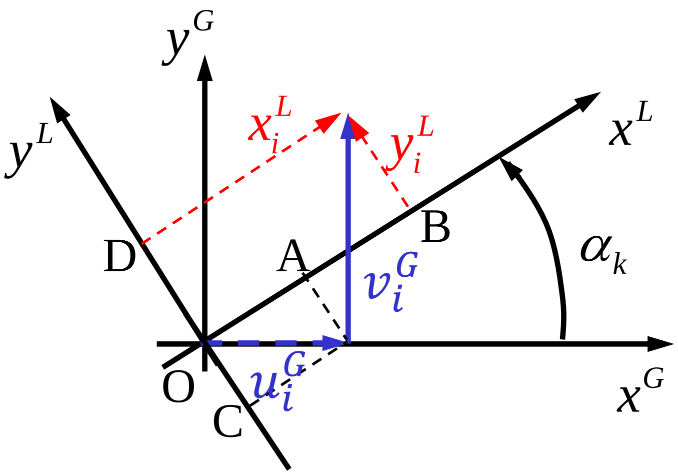
$$= \begin{Bmatrix} \delta u_1^G \\ \delta v_1^G \\ \vdots \\ \delta \vartheta_6^G \end{Bmatrix}^T \begin{Bmatrix} H_1^G \\ V_1^G \\ \vdots \\ M_6^G \end{Bmatrix} = \delta \underline{x}_c^T \underline{R}$$

Step 6: Coordinate transformation from local to global reference

- This step is needed to express all nodal displacements along a common reference
- The transformation is performed based on the following relationship:

$$\underline{x}_k^L = [\Lambda_k]\underline{x}_k^G$$

where $[\Lambda_k]$ is the coordinate transformation matrix corresponding to a plane rotation



$$u_i^L = u_i^G \cos\alpha_k + v_i^G \sin\alpha_k$$
$$v_i^L = -u_i^G \sin\alpha_k + v_i^G \cos\alpha_k$$
$$\vartheta_i^L = \vartheta_i^G$$

$$\Downarrow$$

$$\underline{x}_i^L = [\lambda_k]\underline{x}_i^G \qquad [\lambda_k] = \begin{bmatrix} \cos\alpha_k & \sin\alpha_k & 0 \\ -\sin\alpha_k & \cos\alpha_k & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 6: Coordinate transformation from local to global reference

In the same way for node “j”:

$$\underline{x}_j^L = [\lambda_k]\underline{x}_j^G$$

$$\underline{x}_k^L = \begin{Bmatrix} \underline{x}_i^L \\ \underline{x}_j^L \end{Bmatrix} \; ; \; \underline{x}_k^G = \begin{Bmatrix} \underline{x}_i^G \\ \underline{x}_j^G \end{Bmatrix} \qquad \Rightarrow [\Lambda_k] = \begin{bmatrix} [\lambda_k] & [0] \\ [0] & [\lambda_k] \end{bmatrix}$$

Note: $[\Lambda_k]$ is a constant matrix, as far as the system undergoes small displacements that do not affect the slope in the global reference of the elements, therefore

$$\underline{x}_k^L = [\Lambda_k]\underline{x}_k^G \Rightarrow \underline{\dot{x}}_k^L = [\Lambda_k]\underline{\dot{x}}_k^G$$

Step 6: Coordinate transformation from local to global reference

The kinematical relationship in the previous slide is introduced in the expressions for the kinetic energy, potential energy and virtual work, obtaining the element's mass and stiffness matrices (and forcing vector) in the **global** reference:

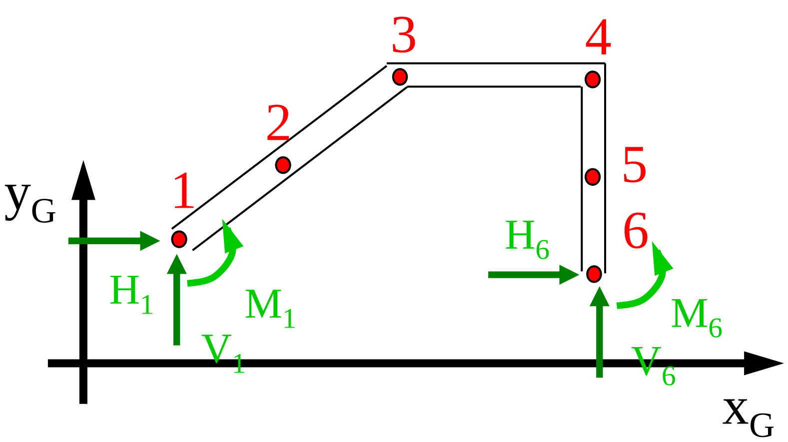
$$T_k = \tfrac{1}{2}\left(\underline{\dot{x}}_k^L\right)^T \left[m_k^L\right]\left(\underline{\dot{x}}_k^L\right) = \tfrac{1}{2}\left(\underline{\dot{x}}_k^G\right)^T \left[\Lambda_k\right]^T \left[m_k^L\right]\left[\Lambda_k\right]\left(\underline{\dot{x}}_k^G\right) = \tfrac{1}{2}\left(\underline{\dot{x}}_k^G\right)^T \left[m_k^G\right]\left(\underline{\dot{x}}_k^G\right)$$

$$V_k = \tfrac{1}{2}\left(\underline{x}_k^L\right)^T \left[k_k^L\right]\left(\underline{x}_k^L\right) = \tfrac{1}{2}\left(\underline{x}_k^G\right)^T \left[\Lambda_k\right]^T \left[k_k^L\right]\left[\Lambda_k\right]\left(\underline{x}_k^G\right) = \tfrac{1}{2}\left(\underline{x}_k^G\right)^T \left[k_k^G\right]\left(\underline{x}_k^G\right)$$

$$\delta L_k = \left(\delta \underline{x}_k^L\right)^T \underline{f}_k^L = \left(\delta \underline{x}_k^G\right)^T \left[\Lambda_k\right]^T \underline{f}_k^L = \left(\delta \underline{x}_k^G\right)^T \underline{f}_k^G$$

Step 7: Numbering of the nodal coordinates and matrix assembling

The nodal displacement components are order into vector $\underline{x}$. The order assigned to the nodal coordinates is arbitrary, but nodal displacements subjected to constraints must be kept separate from the other coordinates



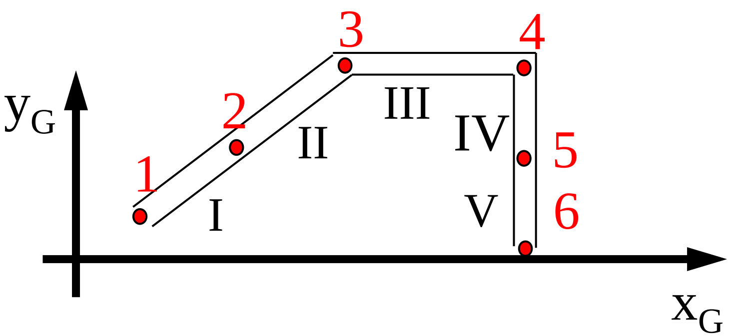
$$\underline{x} = \begin{Bmatrix} u_2^G \\ v_2^G \\ \vartheta_2^G \\ u_3^G \\ \vdots \\ \vartheta_5^G \\ \hline u_1^G \\ \vdots \\ \vartheta_6^G \end{Bmatrix} = \begin{Bmatrix} \{\underline{x}_f\} \\ \{\underline{x}_c\} \end{Bmatrix}$$

Step 7: Matrix assembling

The assembling of the total mass and stiffness matrices may be introduced via the concept of "extraction matrix" $[E_k]$:

$$\underline{x}_k^G = [E_k]\underline{x}$$

Taking as example finite element I:



$$\underline{x} = \begin{Bmatrix} u_2^G \\ v_2^G \\ \vartheta_2^G \\ u_3^G \\ \vdots \\ \vartheta_5^G \\ u_1^G \\ \vdots \\ \vartheta_6^G \end{Bmatrix} \qquad \underline{x}_I^G = \begin{Bmatrix} u_1^G \\ v_1^G \\ \vartheta_1^G \\ u_2^G \\ v_2^G \\ \vartheta_2^G \end{Bmatrix}$$

$$[E_I] = \begin{bmatrix} 0 & 0 & 0 & \cdots & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 7: Matrix assembling

The use of the extraction matrix allows to express the energetic quantities of the k-th element as function of the entire vector x instead than of the element's coordinates $\underline{x}_k^G$ and then to obtain the structural matrices of the entire system

$$T = \sum_k T_k = \sum_k \tfrac{1}{2} \left(\underline{\dot{x}}_k^G\right)^T \left[m_k^G\right] \left(\underline{\dot{x}}_k^G\right) = \tfrac{1}{2} \underline{\dot{x}}^T \underbrace{\sum_k \left(\left[E_k\right]^T \left[m_k^G\right]\left[E_k\right]\right)}_{[m]} \underline{\dot{x}} = \tfrac{1}{2} \underline{\dot{x}}^T \left[m\right] \underline{\dot{x}}$$

By the same procedure:

$$V = \tfrac{1}{2} \underline{x}^T \left[k\right] \underline{x} \quad \text{with} \quad \left[k\right] = \sum_k \left[E_k\right]^T \left[k_k^G\right]\left[E_k\right]$$

$$\delta L = \delta \underline{x}^T \underline{f} \quad \text{with} \quad \underline{f} = \sum_k \left[E_k\right]^T \underline{F}_k^G + \left[E_v\right]^T \underline{R}$$

Outcome of the matrix assembling procedure

Using a block representation of the matrix operation outlined in the last slide:

$$\underline{x} = \begin{Bmatrix} u_2^G \\ v_2^G \\ \vartheta_2^G \\ u_3^G \\ \vdots \\ \vartheta_5^G \\ u_1^G \\ \vdots \\ \vartheta_6^G \end{Bmatrix}$$

$$[E_I] = \begin{bmatrix} [0] & [0] & [0] & [0] & [I] & [0] \\ [I] & [0] & [0] & [0] & [0] & [0] \end{bmatrix} \quad ; \quad [m_I^G] = \begin{bmatrix} [m_{I_{11}}] & [m_{I_{12}}] \\ [m_{I_{21}}] & [m_{I_{22}}] \end{bmatrix}$$

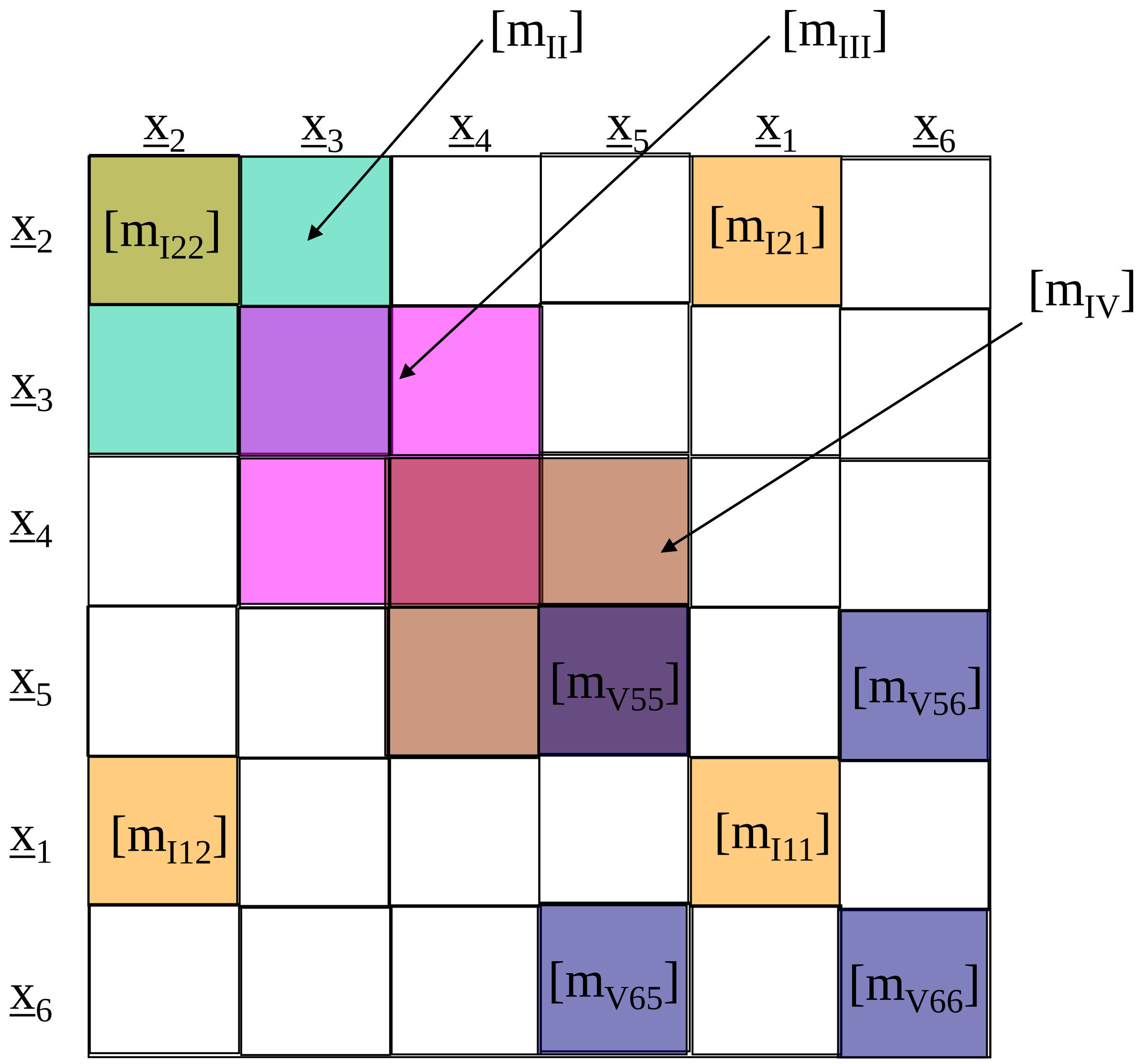
$$[E_I]^T [m_I^G][E_I] = \begin{bmatrix} [m_{I_{22}}] & [0] & [0] & [0] & [m_{I_{21}}] & [0] \\ [0] & [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] & [0] \\ [m_{I_{12}}] & [0] & [0] & [0] & [m_{I_{11}}] & [0] \\ [0] & [0] & [0] & [0] & [0] & [0] \end{bmatrix}$$

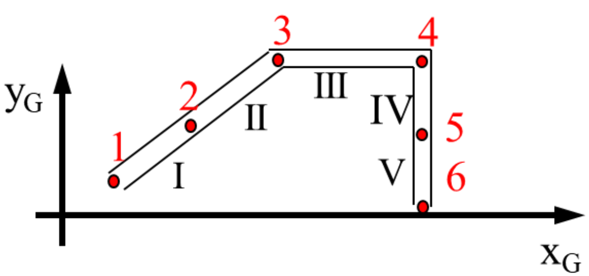
Where we see that the matrix assembling procedure “expands” the element matrix $[m_k^L]$ on a larger matrix having the same dimensions of the total mass matrix [m], filling with zero values the positions that are not corresponding to nodal displacements pertinent to the considered element

Outcome of the matrix assembling procedure

Generalising to all elements, the following structure of the assembled mass matrix is obtained:

$$\underline{x} = \begin{Bmatrix} u_2^G \\ v_2^G \\ \vartheta_2^G \\ u_3^G \\ \vdots \\ \vartheta_5^G \\ u_1^G \\ \vdots \\ \vartheta_6^G \end{Bmatrix}$$





Step 8: Model of structural damping

Assumption: *proportional damping*

$$D = \tfrac{1}{2}\,\underline{\dot{x}}^{T}\left[c\right]\underline{\dot{x}}$$

$$with$$

$$\left[c\right] = \alpha\left[m\right] + \beta\left[k\right]$$

Step 8: Model of structural damping

In order to define α and β values, we observe that by taking a modal approach on the structural matrices of the system, due to the othogonality of the modes of vibration, the modal mass, stiffness and damping matrices will be diagonal, with:

$$\begin{bmatrix} c_1 & & & \\ & c_2 & & \\ & & \ddots & \\ & & & c_n \end{bmatrix} = \alpha \begin{bmatrix} m_1 & & & \\ & m_2 & & \\ & & \ddots & \\ & & & m_n \end{bmatrix} + \beta \begin{bmatrix} k_1 & & & \\ & k_2 & & \\ & & \ddots & \\ & & & k_n \end{bmatrix}$$

With:

$$c_i = \alpha m_i + \beta k_i \qquad \rightarrow \quad \frac{c_i}{c_{i_{cr}}} = \frac{\alpha m_i}{2 m_i \omega_i} + \frac{\beta k_i}{2 m_i \omega_i} = \frac{\alpha}{2\omega_i} + \frac{\beta \omega_i}{2}$$

Step 8: Model of structural damping

If the actual damping factors associated to the structure modes of vibration are known, the following system of equations may be written for "n" modes of interest:

$$\begin{cases} \dfrac{\alpha}{2\omega_1} + \dfrac{\beta\omega_1}{2} = \left(\dfrac{c_1}{c_{1_{cr}}}\right)_{\exp} \\ \cdots \\ \dfrac{\alpha}{2\omega_n} + \dfrac{\beta\omega_n}{2} = \left(\dfrac{c_n}{c_{n_{cr}}}\right)_{\exp} \end{cases}$$

Or, in matrix form:

$$\begin{bmatrix} \dfrac{1}{2\omega_1} & \dfrac{\omega_1}{2} \\ \cdots & \cdots \\ \dfrac{1}{2\omega_n} & \dfrac{\omega_n}{2} \end{bmatrix} \begin{Bmatrix} \alpha \\ \beta \end{Bmatrix} = \begin{Bmatrix} \left(\dfrac{c_1}{c_{1_{cr}}}\right)_{\exp} \\ \cdots \\ \left(\dfrac{c_n}{c_{n_{cr}}}\right)_{\exp} \end{Bmatrix}$$

Which is an over-determined linear system in unwknowns α and β, of the type:

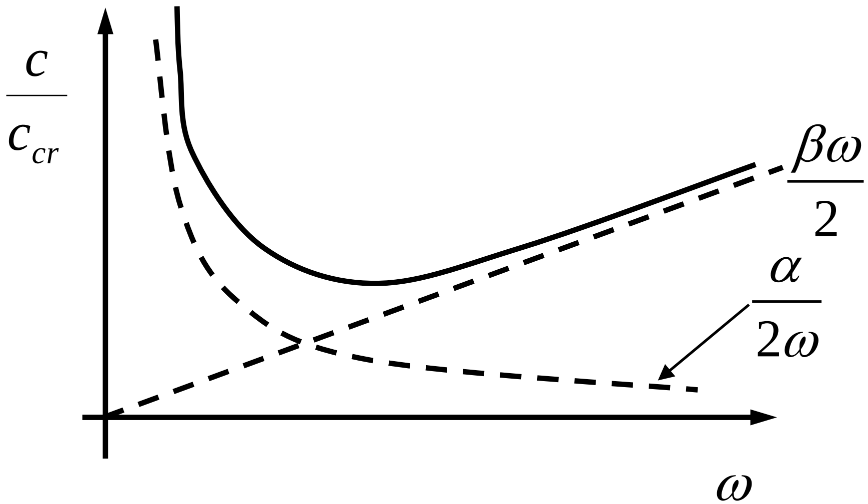
$$[A]\underline{x} = \underline{b}$$

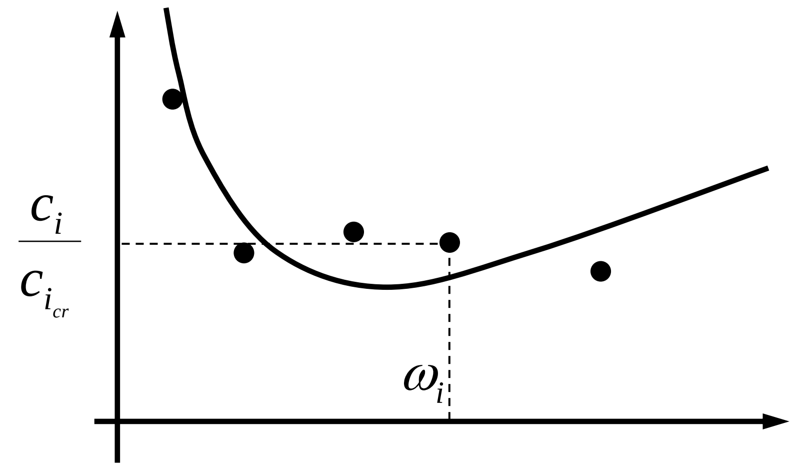
Having a least-square solution provided by:

$$[A]^T[A]\underline{x} = [A]^T\underline{b}$$

Step 8: Model of structural damping

Graphic interpretation of the method to define α and β coefficients





Step 9: Re-introduction of constraints and final equations.

Using the matrix form of Lagrange equations, the following matrix equation is obtained:

$$[m]\ddot{\underline{x}} + [c]\dot{\underline{x}} + [k]\underline{x} = \underline{f}(t)$$

We observe however that in this equation a clear separation of unknowns and known terms is not yet achieved, since:

$$\underline{x} = \begin{Bmatrix} \underline{x}_f \\ \underline{x}_c \end{Bmatrix}$$

unknowns

Known boundary displacements (eventually zero)

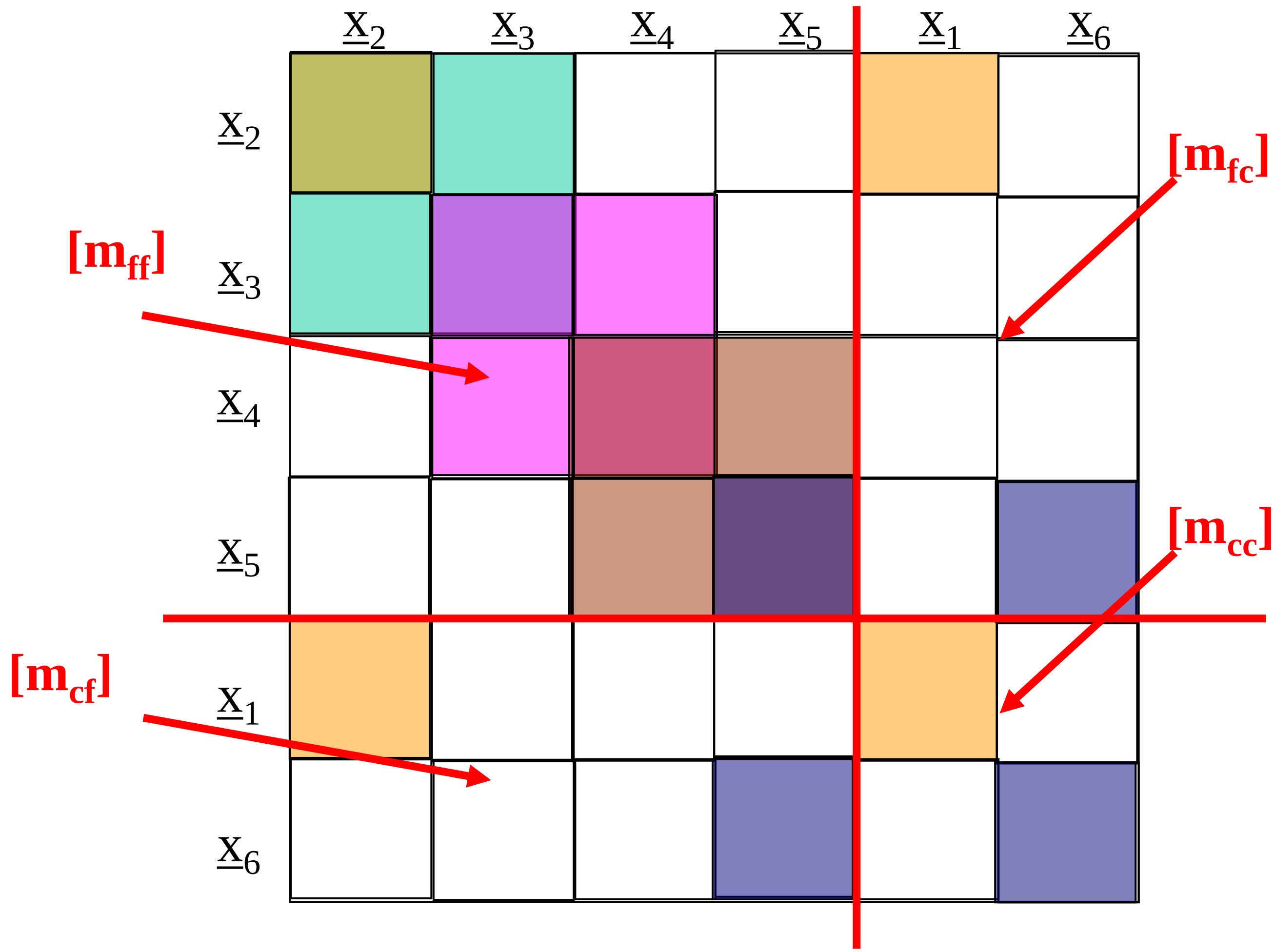
$$\underline{f} = \begin{Bmatrix} \underline{f}_f \\ \underline{f}_c + \underline{R} \end{Bmatrix}$$

Known time dependent forces

Unknown constraint reactions

Step 9: Re-introduction of constraints and final equations.

Therefore, we divide the single matrix equation of the last slide into two separate matrix equations, by partitioning the structural matrices as shown aside:



Step 9: Re-introduction of constraints and final equations.

Using a block representation of matrix products, the single equation obtained before is splitted into the following two equations:

$$\left[m_{ff}\right]\ddot{\underline{x}}_f + \left[m_{fc}\right]\ddot{\underline{x}}_c + \left[c_{ff}\right]\dot{\underline{x}}_f + \left[c_{fc}\right]\dot{\underline{x}}_c + \left[k_{ff}\right]\underline{x}_f + \left[k_{fc}\right]\underline{x}_c = \underline{f}_f$$

$$\left[m_{cf}\right]\ddot{\underline{x}}_f + \left[m_{cc}\right]\ddot{\underline{x}}_c + \left[c_{cf}\right]\dot{\underline{x}}_f + \left[c_{cc}\right]\dot{\underline{x}}_c + \left[k_{cf}\right]\underline{x}_f + \left[k_{cc}\right]\underline{x}_c = \underline{f}_c + \underline{R}$$

In the first of the above equations, the only unknown is vector $\underline{x}_L$ this equation may be revritten in the form: :

$$\left[m_{ff}\right]\ddot{\underline{x}}_f + \left[c_{ff}\right]\dot{\underline{x}}_f + \left[k_{ff}\right]\underline{x}_f = \underline{f}_f - \left(\left[m_{fc}\right]\ddot{\underline{x}}_c + \left[c_{fc}\right]\dot{\underline{x}}_c + \left[k_{fc}\right]\underline{x}_c\right)$$

and solved with respect to $\underline{x}_f$

Once solved this equation, it is possible to replace $\underline{x}_f$ in the second one and use this second equation to derive the unknown constraint reactions $\underline{R}$.

Step 10: Solution of the equations of motion

10.1 the natural frequencies and modes of vibration of the undamped system are provided by the solution of the eigenvalue/eigenvector problem for matrix $[M_{ff}]^{-1}[K_{ff}]$:

$$\left[m_{ff}\right]\ddot{\underline{x}}_f + \left[k_{ff}\right]\underline{x}_f = 0$$

$$\underline{x}_f = \underline{X}_0 e^{i\omega t} \;\; ; \;\; \ddot{\underline{x}}_f = -\omega^2 \underline{X}_0 e^{i\omega t}$$

$$\left(-\omega^2\left[m_{ff}\right] + \left[k_{ff}\right]\right)\underline{X}_0 = 0$$

$$\left(\omega^2\left[I\right] - \left[m_{ff}\right]^{-1}\left[k_{ff}\right]\right)\underline{X}_0 = 0$$

Step 10: Solution of the equations of motion

10.2 the stationary response to a harmonic force is:

$$\left[m_{ff}\right]\underline{\ddot{x}}_f + \left[c_{ff}\right]\underline{\dot{x}}_f + \left[k_{ff}\right]\underline{x}_f = \underline{f}_0 e^{i\Omega t}$$

$$\underline{x}_f = \mathrm{Re}\left(\underline{x}_f^*\right) \; ; \; \underline{x}_f^* = \underline{X}_0 e^{i\Omega t} \; ; \; \underline{\dot{x}}_f^* = i\Omega \underline{X}_0 e^{i\Omega t} \; ; \; \underline{\ddot{x}}_L^* = -\Omega^2 \underline{X}_0 e^{i\Omega t}$$

$$\left(-\Omega^2\left[m_{ff}\right] + i\Omega\left[c_{ff}\right] + \left[k_{ff}\right]\right)\underline{X}_0 = \underline{f}_0$$

$$\underline{X}_0 = \left(-\Omega^2\left[m_{ff}\right] + i\Omega\left[c_{ff}\right] + \left[k_{ff}\right]\right)^{-1}\underline{f}_0 = \left[G(i\Omega)\right]\underline{f}_0$$

And in particular the frequency response function matrix $[G(i\Omega)]$ is:

$$\left[G(i\Omega)\right] = \left(-\Omega^2\left[m_{ff}\right] + i\Omega\left[c_{ff}\right] + \left[k_{ff}\right]\right)^{-1}$$

Step 10: Solution of the equations of motion

10.3 the stationary response to harmonic motion applied at the constraints is:

$$\left[m_{ff}\right]\underline{\ddot{x}}_f+\left[c_{ff}\right]\underline{\dot{x}}_f+\left[k_{ff}\right]\underline{x}_f=-\left(\left[m_{fc}\right]\underline{\ddot{x}}_c+\left[c_{fc}\right]\underline{\dot{x}}_c+\left[k_{fc}\right]\underline{x}_c\right)$$

$$\underline{x}_c=\mathrm{Re}\left(\underline{x}_c^*\right)\ ;\ \underline{x}_c^*=\underline{X}_{c0}e^{i\Omega t}\ ;\ \underline{\dot{x}}_c^*=i\Omega\underline{X}_{c0}e^{i\Omega t}\ ;\ \underline{\ddot{x}}_c^*=-\Omega^2\underline{X}_{c0}e^{i\Omega t}$$

$$\underline{x}_f=\mathrm{Re}\left(\underline{x}_f^*\right)\ ;\ \underline{x}_f^*=\underline{X}_0e^{i\Omega t}\ ;\ \underline{\dot{x}}_f^*=i\Omega\underline{X}_0e^{i\Omega t}\ ;\ \underline{\ddot{x}}_f^*=-\Omega^2\underline{X}_0e^{i\Omega t}$$

$$\left(-\Omega^2\left[m_{ff}\right]+i\Omega\left[c_{ff}\right]+\left[k_{ff}\right]\right)\underline{X}_0=-\left(-\Omega^2\left[m_{fc}\right]+i\Omega\left[c_{fc}\right]+\left[k_{fc}\right]\right)\underline{X}_{c0}$$

$$\underline{X}_0=-\left(-\Omega^2\left[m_{ff}\right]+i\Omega\left[c_{ff}\right]+\left[k_{ff}\right]\right)^{-1}\left(-\Omega^2\left[m_{fc}\right]+i\Omega\left[c_{fc}\right]+\left[k_{fc}\right]\right)\underline{X}_{c0}=$$

$$=\left[G_c(i\Omega)\right]\underline{X}_{c0}$$

And in particular the frequency response function matrix corresponding to constraint motion $[G_c(i\Omega)]$ is:

$$\left[G_c(i\Omega)\right]=-\left(-\Omega^2\left[m_{ff}\right]+i\Omega\left[c_{ff}\right]+\left[k_{ff}\right]\right)^{-1}\left(-\Omega^2\left[m_{fc}\right]+i\Omega\left[c_{fc}\right]+\left[k_{fc}\right]\right)$$